

# CS 486/686

# Independence and Bayesian Networks I

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Lecture 7

RN 12.4, 13.1–13.2 · PM 8.2–8.3

# Reminders

**DUE TONIGHT**

**Tue Jun 2, 11:59 PM**

**Chat 3 (Probabilities) on Chrysalis.**

Search

›

Uncertainty

›

Decisions

›

Learning

## Learning goals

- Tell if two variables are **independent**, or **conditionally independent** given a third
- Derive a **compact representation** of a joint distribution using independence
- Describe the components of a **Bayesian network**
- Compute joint probabilities *from* a Bayes net
- Explain independence in the three key structures

# Why Bayes nets? The $2^n$ problem

A joint distribution over  $n$  Boolean variables needs  $2^n$  probabilities.

| Variables  | Probabilities |
|------------|---------------|
| 6 (Holmes) | 64            |
| 10         | 1,024         |
| 20         | ~ 1 million   |
| 30         | ~ 1 billion   |

**Independence** lets us factor this huge joint into tiny pieces.

# Unconditional independence

**Independence.**  $X$  and  $Y$  are (*unconditionally*) independent iff (any, and therefore all):

$$P(X|Y) = P(X)$$

$$P(Y|X) = P(Y)$$

$$P(X \wedge Y) = P(X) P(Y)$$

Learning  $Y$  does **not** change our belief about  $X$ .

Two probabilities (the priors) suffice to recover four joint probabilities.

# Conditional independence

**Conditional independence.**  $X$  and  $Y$  are *conditionally independent* given  $Z$  iff (any, and therefore all):

$$P(X|Y \wedge Z) = P(X|Z)$$

$$P(Y|X \wedge Z) = P(Y|Z)$$

$$P(X \wedge Y|Z) = P(X|Z) P(Y|Z)$$

Once we know  $Z$ , learning  $Y$  doesn't change our belief about  $X$ .

**Important:** independence does *not* imply conditional independence, and vice versa. We'll see both on later slides.

# Quiz pack: compact representation

Three Boolean variables  $A, B, C$ . How many probabilities specify the joint?

**Q1.** No independence assumed.

7. Chain rule:  $P(A) + P(B|A) + P(C|A \wedge B) = 1 + 2 + 4 = 7$ .

**Q2.**  $A, B, C$  all mutually independent.

3. Just the priors:  $P(A), P(B), P(C) = 3$ .

**Q3.**  $A, B$  conditionally independent given  $C$ .

5.  $P(C) + P(A|C) + P(B|C) = 1 + 2 + 2 = 5$ .

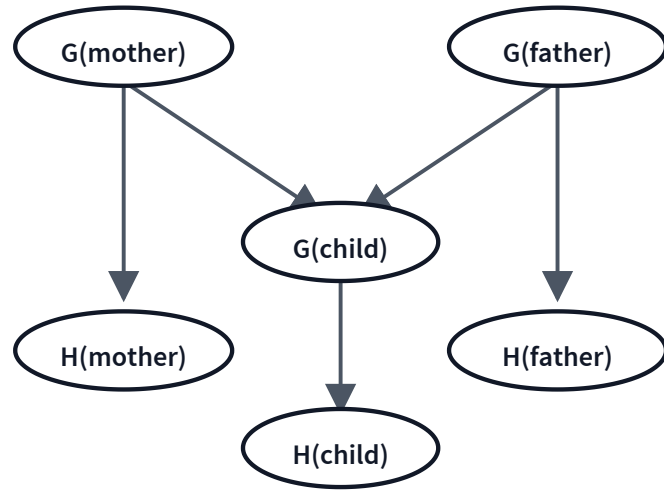
## Bayes net = DAG + CPTs

A **Bayesian network** is a *directed acyclic graph* (DAG) in which:

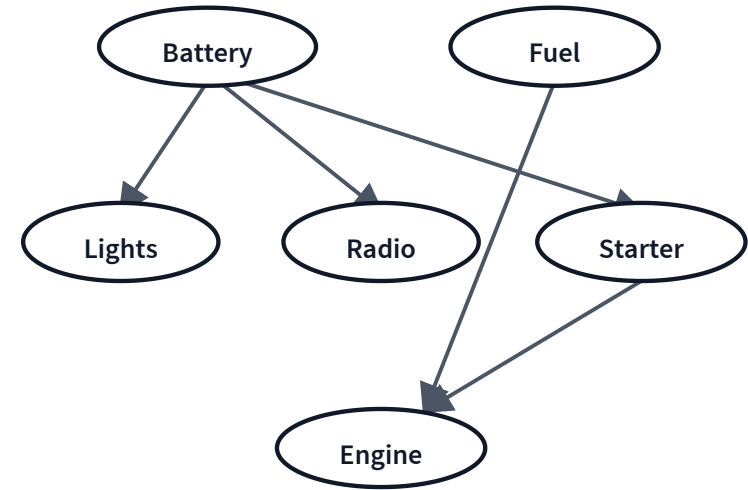
- each node is a random variable,
- each edge  $X \rightarrow Y$  says " $X$  directly affects  $Y$ " —  $X$  is a *parent* of  $Y$ ,
- each node  $X_i$  carries a conditional probability table  $P(X_i \mid \text{Parents}(X_i))$ .

# Examples of Bayes nets

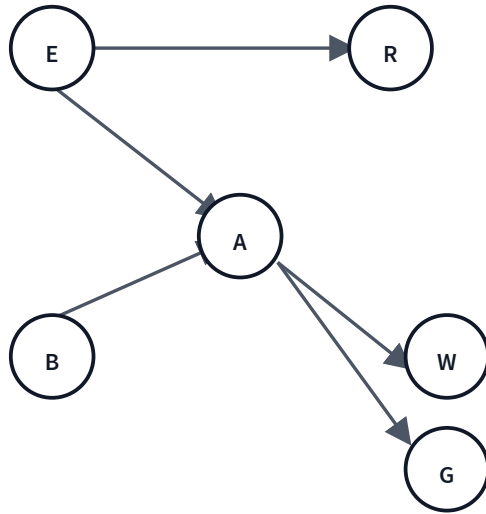
Inheritance of handedness



Car diagnostic network



# The Holmes Bayes net



$$P(E) = 0.0003$$

$$P(B) = 0.0001$$

$$P(R \mid \cdot)$$

$$E: 0.9 \quad \neg E: 0.0002$$

$$P(A \mid \cdot)$$

$$\begin{array}{ll} B \wedge E: 0.96 & B \wedge \neg E: 0.95 \\ \neg B \wedge E: 0.20 & \neg B \wedge \neg E: 0.01 \end{array}$$

$$P(W \mid \cdot)$$

$$A: 0.80 \quad \neg A: 0.40$$

$$P(G \mid \cdot)$$

$$A: 0.40 \quad \neg A: 0.04$$

# From 64 numbers to 12

| Representation                            | Probabilities needed |
|-------------------------------------------|----------------------|
| Full joint over 6 Boolean variables       | $2^6 = 64$           |
| Holmes Bayes net: $1 + 1 + 2 + 4 + 2 + 2$ | 12                   |

From those 12 numbers we can reconstruct every one of the 64 joint probabilities.

For larger networks, the savings are exponential.

**Joint = product of conditionals**

**Bayes-net factorization.** For a Bayes net over variables  $X_1, \dots, X_n$ :

$$P(X_1, \dots, X_n) = \prod_{i=1}^n P(X_i \mid \text{Parents}(X_i)).$$

Each variable is conditionally independent of its non-descendants given its parents.

# Worked example: a joint from the Holmes BN

**Goal:**  $P(\neg B \wedge \neg E \wedge A \wedge \neg R \wedge G \wedge W)$ . All neighbours call but neither a burglary nor an earthquake actually happened.

**Step 1.** Factor by the BN structure (using parents):

$$= P(\neg B) \cdot P(\neg E) \cdot P(A|\neg B \wedge \neg E) \cdot P(\neg R|\neg E) \cdot P(G|A) \cdot P(W|A)$$

**Step 2.** Plug in numbers:

$$\begin{aligned} &= 0.9999 \times 0.9997 \times 0.01 \times 0.9998 \times 0.40 \times 0.80 \\ &\approx \mathbf{3.2 \times 10^{-3}} \end{aligned}$$

## Q: a different all-negatives joint

Q. What is  $P(\neg B \wedge \neg E \wedge \neg A \wedge \neg R \wedge \neg G \wedge \neg W)$ ?

- A. 0.5699
- B. 0.6699
- C. 0.7699
- D. 0.8699
- E. 0.9699

**A — 0.5699.** By the same factorization, each factor is now a "no event happened" version:  
 $= 0.9999 \times 0.9997 \times (1 - 0.01) \times (1 - 0.0002) \times (1 - 0.04) \times (1 - 0.40) \approx 0.5699.$

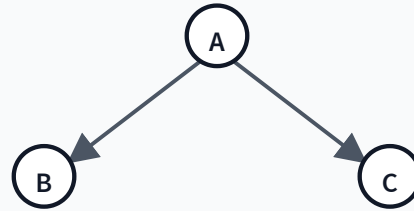
# Three key structures

## Chain



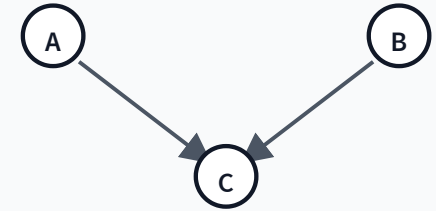
$A$  and  $C$  talk *through*  $B$ .

## Common cause



$B$  and  $C$  share a hidden cause  $A$ .

## Common effect



$A$  and  $B$  both cause  $C$  (a v-structure).

Each structure has its own independence pattern. Let's check.

Chain:  $B \rightarrow A \rightarrow W$



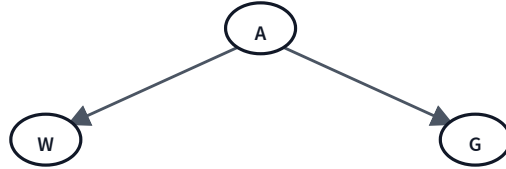
**Q1.** Are  $B$  and  $W$  *unconditionally* independent?

**No.** Knowing  $B$  makes the alarm more likely, which makes  $W$  more likely. Information flows through  $A$ .

**Q2.** Are  $B$  and  $W$  *conditionally* independent given  $A$ ?

**Yes.** Once we know whether the alarm is going,  $W$  only depends on  $A$ ;  $B$  adds nothing more. *The middle node "blocks" the path.*

Common cause:  $W \leftarrow A \rightarrow G$



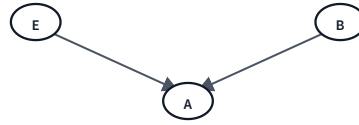
**Q1.** Are  $W$  and  $G$  *unconditionally* independent?

**No.** If  $W$  is calling, the alarm is more likely going, so  $G$  is more likely too. The shared cause  $A$  couples them.

**Q2.** Are  $W$  and  $G$  *conditionally* independent given  $A$ ?

**Yes.** Once  $A$  is known, the two callers are independent observers. *The common cause "blocks" the path when conditioned.*

# Common effect: $E \rightarrow A \leftarrow B$



**Q1.** Are  $E$  and  $B$  *unconditionally* independent?

**Yes.** Earthquakes and burglaries don't influence each other — they only share the same downstream alarm.

**Q2.** Are  $E$  and  $B$  *conditionally* independent given  $A$ ?

**No — the opposite.** Given the alarm, learning of an earthquake makes a burglary less likely: **explaining away**. Conditioning on the common effect *creates* a dependence.

**The v-structure is the surprise:** conditioning on the middle node *opens* a path instead of blocking it.

# The three rules

| Structure                                    | Are endpoints independent? | Given the middle node? |
|----------------------------------------------|----------------------------|------------------------|
| Chain $A \rightarrow B \rightarrow C$        | No                         | Yes <i>blocked</i>     |
| Common cause $B \leftarrow A \rightarrow C$  | No                         | Yes <i>blocked</i>     |
| Common effect $A \rightarrow C \leftarrow B$ | Yes                        | No <i>opens up</i>     |

In the chain and common-cause structures, knowing the middle node *blocks* information flow. In the v-structure, knowing the middle node *opens* it (explaining away).

## Learning goals (recap)

- ✓ Tell if two variables are independent or conditionally independent
- ✓ Derive a compact representation using independence
- ✓ Describe the components of a Bayesian network
- ✓ Compute joint probabilities from a Bayes net
- ✓ Explain independence in the three key structures

## Next: d-separation

The three structures cover paths of length 2. For arbitrary paths in a Bayes net, we generalize to **d-separation** — the rule for reading off all (conditional) independences from any DAG.